

Practice Questions (answers start on a new page)

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CCNA Automation 200-901 v1.1 — Practice Questions

Original items aligned to the official blueprint. These are **not** Cisco exam questions and are not reconstructed from dumps.

How to use: answer Domain by Domain on paper or in a hidden-answer editor. Check yourself in [Answer key](#). Then read the explanation even when you were right.

Legend: **MC** = single answer, **2** = choose two, **INT** = interpretation.

Domain 1 — Software Development and Design (15%)

Q1 (MC). A colleague pastes this snippet. Which format is it, and why would Ansible prefer it over JSON?

```
interfaces:
  - name: GigabitEthernet1
    enabled: true    # management
```

- A. XML, because it supports attributes
- B. JSON, because booleans are lowercase
- C. YAML, because it allows comments and indentation-based structure
- D. YANG, because `enabled` is a leaf

Q2 (MC). Which statement about JSON is true?

- A. Keys may be unquoted if they contain no spaces
- B. Comments start with `#`
- C. `true`, `false`, and `null` are valid literals
- D. A trailing comma after the last object property is required

Q3 (INT). After `data = json.loads('{\"vlans\": [10, 20]}')`, what is `type(data[\"vlans\"])` in Python?

- A. `tuple`
- B. `list`

- C. `str`
- D. `set`

Q4 (MC). You parse XML with `xml.etree.ElementTree` and `find("interface")` returns `None` even though the file contains `<interface>` tags. The root has `xmlns="urn:ietf:params:xml:ns:yang:ietf-interfaces"`. What is the most likely cause?

- A. `ElementTree` cannot parse YANG XML
- B. The default namespace is not included in the search
- C. You must use `yaml.safe_load` on XML
- D. The file is actually JSON

Q5 (MC). In test-driven development, what happens first?

- A. Refactor production code for style
- B. Write a test that fails because the feature does not exist yet
- C. Deploy to production, then write tests from bugs
- D. Generate tests automatically from YANG

Q6 (MC). A 18-month router OS rewrite is fully specified up front, coded, then tested. Which method is this?

- A. Agile
- B. Lean
- C. Waterfall
- D. Observer

Q7 (2). Which two are advantages of putting code in functions and modules? (Choose two.)

- A. The interpreter skips syntax errors
- B. Units can be tested in isolation
- C. The same logic can be reused without copy-paste
- D. JSON parsers run faster

Q8 (MC). A monitoring dashboard updates automatically when a device object changes state, without the device code knowing about the dashboard. Which pattern is this closest to?

- A. Waterfall
- B. MVC View only
- C. Observer
- D. Bare metal

Q9 (MC). In MVC applied to a REST automation app, the HTTP handler that validates input and calls the Meraki client is best described as the:

- A. Model

- B. View
- C. Controller
- D. Observer subject

Q10 (MC). Which is an advantage of version control?

- A. It encrypts REST payloads by default
- B. It provides history, branching, and the ability to revert
- C. It replaces unit tests
- D. It converts XML to JSON

Q11 (MC). `git clone https://github.com/org/lab.git` primarily:

- A. Creates a branch named clone
- B. Copies the repository and its history to a new local directory
- C. Stages all remote files
- D. Opens a pull request

Q12 (MC). You modified `playbook.yml` but `git commit -m "update"` creates a commit with no file changes. What did you forget?

- A. `git push`
- B. `git add playbook.yml`
- C. `git branch`
- D. `git diff --staged` after the commit

Q13 (MC). `git pull` on a tracking branch typically:

- A. Deletes local commits
- B. Fetches remote commits and integrates them into the current branch
- C. Sends commits without fetching
- D. Creates a bare repository

Q14 (INT). A merge conflict shows:

```
<<<<<<< HEAD
hostname: edge-01
=====
hostname: edge-02
>>>>>>> feature
```

What must you do before the merge is complete?

- A. Run `git clone` again
- B. Edit to one correct result, `git add`, then commit the merge

- C. Delete the repository
- D. Only run `git push --force`

Q15 (INT). In this unified diff, what happened to line `url = "https://router/restconf/ ... " ?`

```
--- a/hostname.py
+++ b/hostname.py
@@ -1,6 +1,7 @@
 import requests
-url = "https://router/restconf/data/Cisco-IOS-XE-native:native/hostname"
+HOST = "https://router"
+url = f"{HOST}/restconf/data/Cisco-IOS-XE-native:native/hostname"
```

- A. The URL line was added only
- B. The URL line was removed and replaced by a HOST variable plus a new URL line
- C. The file was deleted
- D. YAML indentation changed

Domain 2 — Understanding and Using APIs (20%)

Q16 (MC). Docs say: `GET /organizations/{orgId}/devices` Header `X-Cisco-Meraki-API-Key` required. Which request is correctly constructed?

- A. POST with a JSON body and no headers
- B. GET to that path with the API key header and no body
- C. PUT replacing the organization
- D. GET with Basic auth only and a `/netconf` path

Q17 (MC). A ChatOps bot needs near-real-time room messages without asking Webex every 2 seconds. Which pattern fits?

- A. SNMP polling
- B. Webhook subscription
- C. Git clone
- D. DHCP snooping

Q18 (2). Which two are common constraints when consuming APIs? (Choose two.)

- A. Rate limiting
- B. OSPF hello timers
- C. Pagination of large collections
- D. Spanning Tree root priority

Q19 (MC). HTTP 401 vs 403:

- A. 401 means the server does not exist; 403 means JSON is invalid
- B. 401 means authentication failed or is missing; 403 means the identity is known but not allowed
- C. They are interchangeable
- D. 403 only applies to SOAP

Q20 (MC). You send `Content-Type: text/plain` with a JSON body to an API that requires `application/json`. Which status is most typical?

- A. 201
- B. 204
- C. 415
- D. 301

Q21 (MC). Response includes `429` and `Retry-After: 30`. What should a client do?

- A. Immediately retry in a tight loop
- B. Wait at least 30 seconds, then retry; consider backoff
- C. Switch the method from GET to DELETE
- D. Disable TLS

Q22 (INT). Given:

```
HTTP/1.1 201 Created
Location: /networks/N_123
Content-Type: application/json

{"id": "N_123", "name": "Branch"}
```

Which statement is true?

- A. The call failed authentication
- B. A resource was created; the body and Location identify it
- C. This is a NETCONF hello
- D. The client must use port 830

Q23 (MC). Meraki Dashboard API key is sent as:

- A. `Authorization: Basic <key>`
- B. A cookie named `JSESSIONID` only
- C. Header `X-Cisco-Meraki-API-Key`
- D. YANG namespace

Q24 (MC). Catalyst Center token workflow is:

- A. SSH to port 22, then SNMP
- B. POST `/dna/system/api/v1/auth/token` then send `X-Auth-Token` on later calls
- C. Only API keys in query strings
- D. NETCONF hello on 443

Q25 (MC). Which pair is most accurate?

- A. REST uses resource URLs and HTTP verbs; RPC emphasizes calling a named procedure
- B. REST always uses XML; RPC always uses YAML
- C. Asynchronous APIs never return HTTP 202
- D. Webhooks are a form of SNMP trap over UDP 161

Q26 (INT). What does this script do?

```
import requests
r = requests.get(
    "https://httpbin.org/get",
    params={"device": "r1"},
    headers={"Accept": "application/json"},
    timeout=10,
)
print(r.status_code, r.json()["args"])
```

- A. NETCONF `<edit-config>`
- B. REST GET with a query parameter, then prints status and echoed args
- C. Docker build
- D. Ansible `gather_facts`

Q27 (MC). `requests.post(url, json={"hostname": "r1"})` sets:

- A. Form-urlencoded body and no Content-Type
- B. A JSON body and Content-Type `application/json`
- C. A YAML playbook
- D. An XML RPC

Q28 (MC). A job API returns `202 Accepted` and a URL `/tasks/88`. The client later GETs that URL. This is:

- A. Synchronous RPC completing in one round trip
 - B. An asynchronous pattern
 - C. A Git merge conflict
 - D. CSRF
-

Domain 3 — Cisco Platforms and Development (15%)

Q29 (MC). You must list wireless APs in a cloud-managed small-branch org with a Dashboard API key. Which platform?

- A. UCS Manager
- B. Meraki
- C. CUCM AXL
- D. Firepower Management Center only

Q30 (MC). `/dna/intent/api/v1/network-device` after a token fetch is characteristic of:

- A. Meraki
- B. Cisco Catalyst Center
- C. Webex Meetings
- D. NTP

Q31 (MC). Cisco ACI programmatic access is primarily through:

- A. The APIC object model / REST
- B. Meraki organization inventory
- C. Webex `/rooms`
- D. Docker Hub

Q32 (MC). Overlay WAN policies, device templates, and vManage REST are capabilities of:

- A. UCS Manager
- B. Cisco Catalyst SD-WAN
- C. ISE pxGrid only
- D. GitHub Actions

Q33 (MC). NSO is best described as:

- A. A wireless LAN controller CLI
- B. A service orchestrator that maps service models to device models
- C. A Python unit-test runner
- D. An OWASP scanner

Q34 (2). Which two are compute management platforms on the blueprint? (Choose two.)

- A. UCS Manager
- B. Intersight
- C. Webex
- D. Secure Malware Analytics

Q35 (MC). CUCM AXL vs UDS:

- A. AXL is user-facing REST; UDS is SOAP admin
- B. AXL is administrative SOAP/XML; UDS is user-oriented data services
- C. Both are NETCONF on port 830
- D. Both are Meraki API keys

Q36 (MC). You want to create a Webex space, add a person, and post a message. Which resources?

- A. `/rooms` , `/memberships` , `/messages`
- B. `/dna/intent/api/v1/network-device`
- C. `<get-config>`
- D. `docker exec`

Q37 (MC). Identity-based network access policy and session directory APIs point to:

- A. Terraform Cloud
- B. Cisco ISE
- C. NTP
- D. CML

Q38 (MC). You need a live IOS XE box for RESTCONF practice. Which DevNet resource?

- A. Code Exchange only
- B. Sandbox
- C. A unified diff
- D. OWASP Top 10

Q39 (MC). You need sample Python for SD-WAN device lists, not live gear. Which resource?

- A. Sandbox reservation of a 40-node fabric
- B. Code Exchange (or GitHub samples)
- C. TAC severity 1
- D. SNMP trap receiver

Q40 (MC). NETCONF on IOS XE commonly uses:

- A. TCP 830 over SSH, XML RPCs
- B. UDP 161
- C. TCP 23
- D. YAML over Telnet

Q41 (MC). RESTCONF GET for hostname often looks like:

- A. `ssh router show run | json`

- B. GET /restconf/data/Cisco-IOS-XE-native:native/hostname with Accept: application/yang-data+json
- C. POST /webhooks
- D. git pull

Q42 (INT). A YANG `list interface { key "name"; leaf name { type string; } leaf enabled { type boolean; } }` means:

- A. Interfaces are identified by `name`; `enabled` is a boolean leaf
- B. Only XML comments are allowed
- C. The list cannot be queried by RESTCONF
- D. Port 80 is required

Q43 (MC). NX-OS **NX-API CLI** is typically:

- A. JSON-RPC style posting of CLI commands over HTTP(S)
- B. Only SNMP
- C. Only Webex
- D. Only Terraform HCL

Domain 4 — Application Deployment and Security (15%)

Q44 (MC). A benefit of edge computing is:

- A. Higher WAN round-trip for every sensor reading
- B. Processing closer to users or devices to cut latency and backhaul
- C. Replacing DNS
- D. Disabling TLS

Q45 (MC). An app runs in a customer-owned data center on OpenStack, with burst VMs in AWS. This is:

- A. Public cloud only
- B. Private cloud only
- C. Hybrid cloud
- D. Bare metal exclusive

Q46 (2). Which two are container attributes vs VMs? (Choose two.)

- A. Share the host kernel
- B. Each container must include a full guest OS kernel
- C. Images are typically thinner than VM images

- D. They cannot run in public cloud

Q47 (MC). In a CI/CD pipeline, unit tests that run on every commit belong mainly to:

- A. Manual change-advisory meetings only
- B. Continuous integration
- C. Spanning Tree
- D. NAT overload

Q48 (INT). Which test is correctly constructed?

```
import unittest
from net import mask_to_prefix
class T(unittest.TestCase):
    def test_24(self):
        self.assertEqual(mask_to_prefix("255.255.255.0"), 24)
```

- A. It is invalid because unittest cannot test functions
- B. It compares expected 24 to the function result
- C. It starts a Docker daemon
- D. It sends NETCONF

Q49 (INT). In a Dockerfile, `CMD ["python", "app.py"]` runs:

- A. At image **build** time
- B. As the default process when a container **starts**
- C. Only on Windows hosts
- D. Instead of FROM

Q50 (MC). `EXPOSE 8080` in a Dockerfile:

- A. Publishes the port to the host by itself
- B. Documents the intended port; publishing still needs `-p` (or compose)
- C. Opens 8080 on the corporate firewall
- D. Enables NETCONF

Q51 (MC). Storing `MERAKI_API_KEY` in a public GitHub repo is primarily a failure of:

- A. Secret protection
- B. VLAN pruning
- C. NTP stratum
- D. MVC View

Q52 (MC). TLS for HTTPS API calls is:

- A. Encryption at rest
- B. Encryption in transit
- C. CSRF
- D. A YANG list key

Q53 (MC). A reverse proxy in front of microservices typically:

- A. Replaces MAC learning in a switch ASIC
- B. Terminates TLS and routes by path or hostname to backends
- C. Assigns DHCP leases
- D. Runs OSPF

Q54 (MC). SQL injection is best described as:

- A. Injecting script into HTML rendered for other users
- B. Inserting attacker-controlled SQL via unsanitized input
- C. Forging a request from a victim's browser to another site
- D. Rate limiting

Q55 (MC). XSS is:

- A. Cross-site scripting in the browser
- B. XML Site Security
- C. A NETCONF capability
- D. A Docker layer

Q56 (MC). CSRF:

- A. Is the same as SSH brute force
- B. Tricks an authenticated browser into sending an unwanted state-changing request
- C. Is a YAML indent error
- D. Is encryption at rest

Q57 (INT). What does this Bash fragment do?

```
mkdir -p /opt/lab && cd /opt/lab
export APP_ENV=prod
cp /tmp/app.env .env
```

- A. Creates a directory, changes into it, sets an environment variable, copies a file
- B. Starts nginx
- C. Commits to Git
- D. Builds a Docker image

Q58 (MC). DevOps principles emphasize:

- A. Isolated teams throwing releases over a wall with no telemetry
 - B. Collaboration, automation, measurement, and sharing
 - C. Telnet as the only management plane
 - D. Disabling version control
-

Domain 5 — Infrastructure and Automation (20%)

Q59 (MC). Model-driven programmability adds value because:

- A. Operators scrape always-changing CLI text with no schema
- B. YANG models define structured, validatable data that NETCONF/RESTCONF can read and write
- C. It eliminates the need for authentication
- D. It only works with Telnet

Q60 (MC). Configuring one switch via RESTCONF is **device-level**. Pushing campus intent to hundreds of devices via Catalyst Center is:

- A. Also device-level only
- B. Controller-level management
- C. CSRF
- D. Bare metal

Q61 (2). Which two tool roles match the blueprint? (Choose two.)

- A. Cisco Modeling Labs simulates network topologies
- B. pyATS tests and parses operational state
- C. OWASP is a routing protocol
- D. NTP forwards Ethernet frames

Q62 (MC). Infrastructure as code means:

- A. Clicking uniquely in GUIs so no two devices match
- B. Desired state stored in versioned files and applied repeatably
- C. Disabling Git
- D. Using only Telnet macros

Q63 (MC). Terraform is distinctive among the three blueprint tools because it:

- A. Has no state file
- B. Uses declarative HCL and a state file to plan/apply resources

- C. Is a YANG compiler
- D. Replaces DNS

Q64 (INT). Identify the workflow:

```
token = requests.post(f"https://{host}/dna/system/api/v1/auth/token", auth=(u, p),
verify=False).json()["Token"]
devs = requests.get(
    f"https://{host}/dna/intent/api/v1/network-device",
    headers={"X-Auth-Token": token},
    verify=False,
).json()
```

- A. Meraki client listing
- B. Catalyst Center authentication then device inventory
- C. Webex message send
- D. Docker image pull

Q65 (INT). Identify the workflow:

```
requests.get(
    "https://api.meraki.com/api/v1/organizations",
    headers={"X-Cisco-Meraki-API-Key": key},
)
```

- A. List Meraki organizations
- B. APIC aaalogin
- C. NETCONF commit
- D. Terraform destroy

Q66 (INT). Identify the workflow:

```
requests.get(
    f"https://{host}/restconf/data/ietf-interfaces:interfaces",
    auth=(u, p),
    headers={"Accept": "application/yang-data+json"},
    verify=False,
)
```

- A. RESTCONF read of IETF interfaces
- B. Git merge
- C. CSRF attack
- D. DHCP discover

Q67 (INT). This playbook task:

```
- name: Ensure nginx is installed
  ansible.builtin.package:
    name: nginx
    state: present
```

- A. Deletes nginx
- B. Installs/ensures the nginx package is present (idempotent package management)
- C. Creates a VLAN
- D. Opens port 830

Q68 (INT). Combined with `user: { name: app, state: present }` and `service: { name: nginx, state: started, enabled: true }`, the playbook is automating:

- A. Package install, user management, and service start/enable
- B. Only BGP
- C. Only Docker layer caching
- D. Only Webex rooms

Q69 (INT). Bash:

```
sudo apt-get update
sudo apt-get install -y python3-pip
sudo useradd -m cicd
cd /opt && tar xzf app.tgz
```

- A. File extract plus package and user creation on a Linux host
- B. NETCONF edit-config
- C. A Terraform provider
- D. MVC View rendering

Q70 (INT). RESTCONF body:

```
{ "Cisco-IOS-XE-native:hostname": "Cat8K" }
```

Status 200. What did you learn?

- A. The device hostname is Cat8K
- B. Authentication failed
- C. The path was a webhook
- D. YAML comments were required

Q71 (INT). NETCONF reply contains `<rpc-error>` and `invalid-value`. This means:

- A. Success `ok`

- B. The RPC failed validation or content was rejected
- C. Git conflict
- D. HTTP 201

Q72 (INT). `container interfaces-state { config false; ... leaf oper-status ... }` means:

- A. `oper-status` is writable running-config
- B. Operational (read-only) state, not configuration
- C. A Docker CMD
- D. A VLAN ID

Q73 (MC). A code review process primarily:

- A. Replaces unit tests and monitoring
- B. Lets peers catch bugs, secret leaks, and design issues before merge
- C. Compiles YANG into OSPF
- D. Assigns MAC addresses

Q74 (INT). Sequence: Client → POST `/token` → Controller → 200 token → Client → GET `/devices` → Controller → 200 list. This diagram shows:

- A. Token auth then a synchronous inventory GET
- B. A webhook from device to client first
- C. A merge conflict
- D. DHCP only

Domain 6 — Network Fundamentals (15%)

Q75 (MC). A VLAN's primary purpose is:

- A. Encrypting Git repos
- B. Segmenting a LAN into separate L2 broadcast domains
- C. Replacing DNS
- D. Storing Terraform state

Q76 (MC). A host `192.168.10.10/24` with gateway `192.168.10.1` sends to `8.8.8.8`. The first hop is:

- A. Direct L2 to 8.8.8.8
- B. The default gateway 192.168.10.1
- C. TCP port 830
- D. The host's MAC as destination IP

Q77 (MC). /30 IPv4 prefix typically provides how many usable host addresses on a point-to-point link (classic usable-host counting)?

- A. 254
- B. 2
- C. 16
- D. 1

Q78 (MC). A load balancer's job in a topology is to:

- A. Learn MAC tables only
- B. Distribute client connections across a pool and health-check members
- C. Assign VLANs to Git branches
- D. Run YANG compilers

Q79 (INT). Diagram: Client:443 → firewall:443 → LB:443 → app:8080. Which statement is correct?

- A. The client must open NETCONF 830 to the app
- B. HTTPS is permitted to the LB; the app listens on 8080 internally
- C. Telnet is required
- D. The firewall is a hypervisor

Q80 (MC). OSPF adjacency formation is primarily which plane?

- A. Management plane
- B. Control plane
- C. Data plane only
- D. CI/CD plane

Q81 (MC). SSH to the device CLI uses which plane?

- A. Data plane
- B. Management plane
- C. Spanning Tree plane
- D. Overlay only

Q82 (MC). DHCP provides:

- A. Time sync
- B. Dynamic IPv4/IPv6 address and typically mask, gateway, DNS
- C. Git tags
- D. Dockerfile layers

Q83 (MC). DNS failure with a hardcoded-name API client typically causes:

- A. Faster RESTCONF
- B. Inability to resolve the API hostname to an IP
- C. Automatic YANG compile
- D. VLAN 1 deletion

Q84 (2). Which two port mappings are correct? (Choose two.)

- A. SSH 22/tcp
- B. NETCONF 830/tcp
- C. HTTPS 23/tcp
- D. Telnet 443/tcp

Q85 (MC). RESTCONF on IOS XE is most often reached on:

- A. UDP 123
- B. TCP 443
- C. UDP 67
- D. TCP 23

Q86 (MC). Packet capture shows SYN to `api.corp:443` with no SYN-ACK. Inside users can ping the API host. Most likely:

- A. DNS is wrong (ping used the name successfully)
- B. Transport port 443 is blocked
- C. The client lacks a MAC address
- D. YAML indent

Q87 (MC). The app logs client IP `10.0.255.2` for all users behind a VIP. Users actually sit in `203.0.113.0/24`. Likely:

- A. Source NAT on the firewall or load balancer
- B. A Git rebase
- C. TDD
- D. Observer pattern

Q88 (MC). An HTTPS client works at home but fails on corporate Wi-Fi unless `HTTPS_PROXY` is set. Likely:

- A. Missing proxy configuration (and possibly TLS intercept trust)
- B. NETCONF port closed
- C. Docker EXPOSE
- D. MVC Model

Q89 (MC). Site-to-site VPN is up but traffic to `10.10.20.0/24` never encrypts. Both sides use `10.10.20.0/24` locally. Likely:

- A. Overlapping encryption domains / VPN overlap
- B. JSON trailing comma only
- C. Python GIL
- D. Unit test discovery

Q90 (MC). An API client times out after 2s on a 250 ms RTT satellite link, but curl with a 30s timeout works.

This illustrates:

- A. Application sensitivity to latency/timeouts as a network constraint
 - B. That 401 is always wrong
 - C. That VLANs replace TCP
 - D. That YANG is a transport
-

Answer key

Do not read this section until you have answered.

Q1 C. YAML: indentation, `-` lists, `#` comments. Ansible playbooks are YAML. YANG is a model, not this snippet's encoding.

Q2 C. JSON literals are lowercase `true` / `false` / `null` . Keys must be double-quoted. No comments. Trailing commas are illegal.

Q3 B. JSON arrays become Python `list` .

Q4 B. Default xmlns means unprefixes `find("interface")` misses nodes; use a namespace map.

Q5 B. Red-green-refactor starts with a failing test.

Q6 C. Big-bang specify-then-build is waterfall.

Q7 B and C. Testability and reuse. Functions do not skip syntax errors or speed JSON parsing by themselves.

Q8 C. Observer: dependents are notified of state changes.

Q9 C. Controller maps input to model operations.

Q10 B. History, branches, revert/collaboration.

Q11 B. Clone copies the repo locally.

Q12 B. Commit records the index; you must `git add` first.

Q13 B. Pull fetches and integrates.

Q14 B. Resolve markers, stage, commit.

Q15 B. Minus line removed; two plus lines added (`HOST` and new `url`).

Q16 B. Docs specify GET + API key header.

Q17 B. Webhooks push events; polling is the inefficient alternative.

Q18 A and C. Rate limits and pagination are API consumer constraints. OSPF/STP are not.

Q19 B. 401 authentication; 403 authorization.

Q20 C. 415 unsupported media type.

Q21 B. Honor Retry-After / backoff. Tight loops worsen 429.

Q22 B. 201 + Location + JSON body = created resource.

Q23 C. Meraki uses `X-Cisco-Meraki-API-Key` .

Q24 B. Token endpoint then `X-Auth-Token` . Paths still say `/dna/` .

Q25 A. REST vs RPC is about resources vs procedures.

Q26 B. `requests.get` with `params` is a REST GET.

Q27 B. `json=` serializes and sets Content-Type.

Q28 B. 202 + later GET is async job pattern.

Q29 B. Meraki cloud dashboard API.

Q30 B. Catalyst Center intent API.

Q31 A. APIC object model.

Q32 B. Catalyst SD-WAN / vManage.

Q33 B. NSO service orchestration.

Q34 A and B. UCS Manager and Intersight.

Q35 B. AXL admin SOAP; UDS user services.

Q36 A. Webex rooms, memberships, messages.

Q37 B. ISE.

Q38 B. Sandbox for live gear.

Q39 B. Code Exchange for samples.

Q40 A. NETCONF 830/ssh XML.

Q41 B. RESTCONF yang-data+json path.

Q42 A. list + key + boolean leaf.

Q43 A. NX-API CLI JSON-RPC.

Q44 B. Edge: locality, latency, backhaul.

Q45 C. Hybrid = private + public.

Q46 A and C. Shared kernel, smaller images. B is a VM trait.

Q47 B. CI runs tests on integrate/commit.

Q48 B. Standard TestCase assertion.

Q49 B. CMD is runtime, not build (that's RUN).

Q50 B. EXPOSE is documentation; `-p` publishes.

Q51 A. Secrets in Git.

Q52 B. TLS = in transit.

Q53 B. Reverse proxy TLS/path routing.

Q54 B. SQLi.

Q55 A. XSS.

Q56 B. CSRF.

Q57 A. mkdir/cd/export/cp.

Q58 B. CALMS-style DevOps.

Q59 B. Structured YANG vs CLI scraping.

Q60 B. Controller-level.

Q61 A and B. CML simulates; pyATS tests.

Q62 B. Versioned desired state.

Q63 B. HCL + state + plan/apply.

Q64 B. Catalyst Center token + devices.

Q65 A. Meraki orgs GET.

Q66 A. RESTCONF interfaces.

Q67 B. package state present.

Q68 A. Matches 5.8 verbs: packages, users, service start.

Q69 A. apt, useradd, tar — 5.9.

Q70 A. Hostname native container value.

Q71 B. rpc-error is failure.

Q72 B. config false = operational.

Q73 B. Peer review before merge.

Q74 A. Sync token then GET.

Q75 B. VLAN = L2 segment.

Q76 B. Off-subnet → gateway.

Q77 B. /30 → 2 usable (classic).

Q78 B. LB distribution + health.

Q79 B. 443 to LB, 8080 backend.

Q80 B. Routing protocols = control plane.

Q81 B. SSH = management plane.

Q82 B. DHCP addressing options.

Q83 B. Name resolution failure.

Q84 A and B. 22 and 830. HTTPS is 443, Telnet 23.

Q85 B. RESTCONF HTTPS 443.

Q86 B. Ping ≠ TCP 443 allowed.

Q87 A. SNAT hides real clients.

Q88 A. Explicit proxy requirement.

Q89 A. Overlapping VPN subnets.

Q90 A. Timeouts vs RTT — 6.9.

If you missed more than a couple in a domain, reopen that domain in the complete study guide and re-do the lab before taking a commercial practice exam.